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In [9]: import numpy as np
```

```
In [10]: def mse(y_true: float, y_pred: float) -> float:
          return np.mean(np.square(y_true - y_pred))
          def binary_cross_entropy(y_true: float, y_pred: float) -> float:
              return -(y_true * np.log(y_pred) + (1 - y_true) * np.log(1 - y_pred))
          def gradient_descent(w: float, n: float, dw: float) -> float:
              return w - (n * dw)
```

Task 1

```
In [11]: y_true, y_pred = 1, 0.6
```

```
mean_sq_error = mse(y_true, y_pred)
bce = binary_cross_entropy(y_true, y_pred)

print(f" Mean Squared Error = {mean_sq_error:.4f}")
print(f" BinaryCross Entrpy = {bce:.4f}")
```

Mean Squared Error = 0.1600

BinaryCross Entrpy = 0.5108

Task 2

```
In [12]: w, dw = 0.8, 0.5
          n1, n2 = 0.1, 0.5

          w1 = gradient_descent(w, n1, dw)
          w2 = gradient_descent(w, n2, dw)

          print(f"Updated Weight (n = 0.1) = {w1}")
          print(f"Updated Weight (n = 0.5) = {w2}")
```

Updated Weight ($n = 0.1$) = 0.75

Updated Weight ($n = 0.5$) = 0.55